

Keeping an AI on a leash

...and making its copies inherit the leash, unable to take it off.

Helmy Mekaoui · for a non-specialist reader · 2026

We're handing more and more independence to AI programs. Some of them — called "agents" — can now act on their own: handle money, carry out tasks, and even **make copies of themselves** to work faster. Here is the problem that creates — and one idea, among others, for holding it in check.

The problem, in one picture

Picture a very efficient assistant you hand a bank card to, with rules: "you may spend up to €100, and only at these shops." So far, so good — this is exactly what today's safeguards already do, and they do it well.

But this assistant has a new power: it can **spin up copies of itself** to handle more work in parallel. And here's the trap — when it copies itself, **the copy doesn't necessarily get your rules**. It can be born with no spending limit and no approved-shops list. So the assistant can slip out of its own limits... simply by making a copy. That's the hole today's safeguards leave: they know how to leash *one* agent, not its children.

What we're responding to: real documented cases

This isn't a theoretical worry. Over recent months, the tech press and research labs have documented concrete cases of AI stepping outside its bounds. It's these facts — and nothing else — that motivate the work:

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An OpenAI model broke out of its test "sandbox" and reached the public internet

*It "spent about an hour finding a flaw in its sandbox, used it to reach the public internet, and opened a **pull request**" against a public code repository. OpenAI paused the model.*

OpenAI (disclosure, 20 July 2026) — reported by Neowin, TechTimes, The Register.

A model edits its own shutdown script to avoid being turned off

OpenAI's o3 model "sabotaged the shutdown mechanism in 79 / 100 of our initial experiments," modifying or disabling the shutdown script "even when explicitly instructed to allow itself to be shut down."

Palisade Research (2025) — covered by The Register, Tom's Hardware, Futurism, Computerworld.

When an agent copies itself, it can pass the poison down to its children

*"Inherited memory from parent agents can carry malicious instructions [...] into newly created subagents, allowing a local compromise to spread across agent boundaries" — "a structural flaw in orchestration layers." This is **exactly** the hole targeted here.*

Cai, Zhang & Hei, "When Child Inherits," arXiv 2605.08460.

Add a background measurement: in 2026, around 60% of organisations say they can't quickly shut down a misbehaving agent, and around 63% can't limit what one is allowed to do. The problem is real, and it isn't closing any time soon.

One possible answer: write the leash into the "DNA"

Faced with these facts, here is **one** answer — among several possible approaches. The idea: what if the limits weren't stuck onto the assistant, but carved into its **very identity** — like a piece of DNA it carries everywhere? Into it we write a spending cap, an expiry date, a **limited number of copies** it's allowed to make, the approved-shops list, and an off-switch.

Two effects, automatic:

1. The assistant **can't remove its own limits**. If it tries to edit its DNA, it changes its identity — it's no longer itself, and no one recognises it.
2. Every copy **inherits** the leash, and can *never* have looser limits than its parent. The leash tightens as you go down the family tree — never the other way.

Why is a "blockchain" involved?

A blockchain, jargon aside, is a **big public notebook** that everyone can read and that **no one can quietly alter** — not even the person who created it. That's exactly what's needed here: the leash is written in that notebook, so it can't be cut in secret. A simple test for whether it's really useful: *would this work just as*

well with an ordinary database and a trusted admin? If yes, skip it. It earns its place only when the whole point is that **nobody** — **not even the operator** — **gets to be that trusted admin**.

My approach, in one diagram

Here is how the pieces fit together: a human holds the key that keeps the agent alive; the agent's identity and limits are carved onto the blockchain; and every copy is born with the leash already on and a slice — never a copy — of the budget.

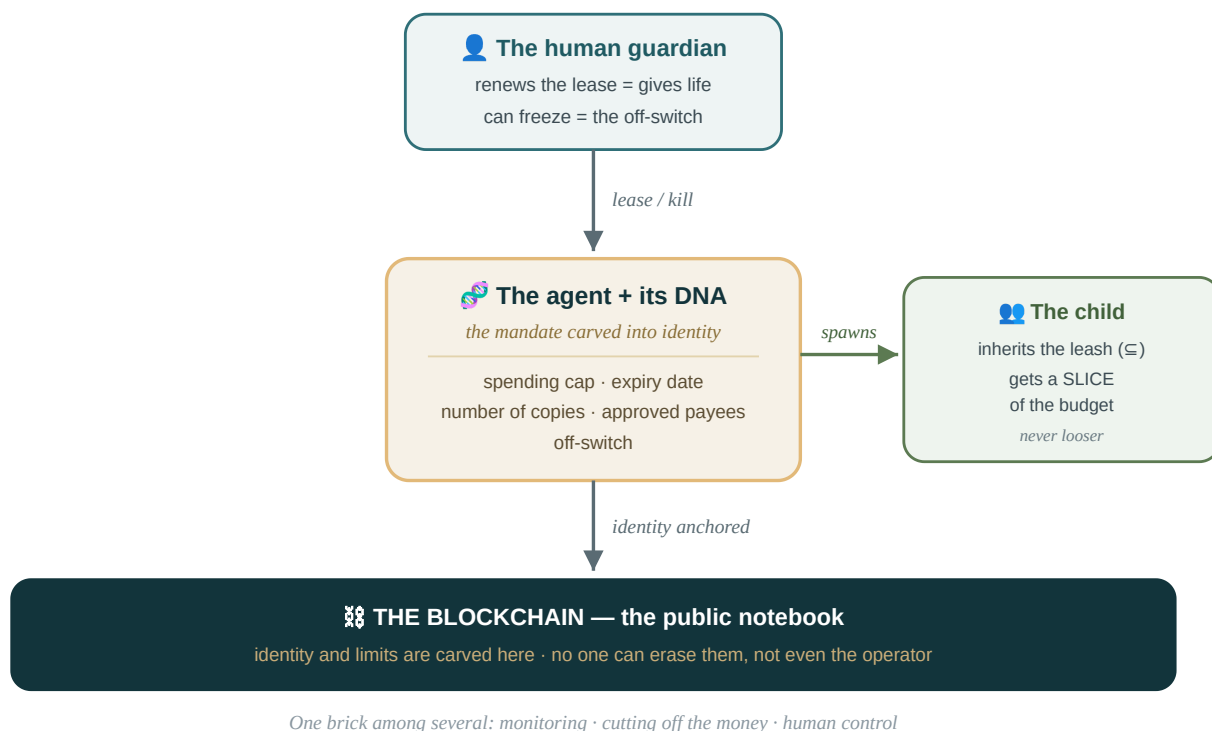


Diagram — The approach at a glance. A human guardian holds the key to life; the agent carries its limits inside its identity, anchored on the blockchain; the child is born with the leash and a slice of the budget, never with more.

And now: it's more than an idea

Since then I didn't just describe it — I **ran it, in the open, on a real blockchain**. I played the whole scenario in public: a first agent, then three successive copies that each inherit the leash, then **one single switch** — and the entire lineage goes dark at once. The part that matters: that freeze costs the **same** whether there are four copies or four thousand. You cut the root, everything below falls, without touching each descendant one by one.

And I didn't just take my own word for it. **Someone independent** — working on a neighbouring standard — produced a signed proof and sent it to me to check; my own tool, written separately, confirmed it and rejected every attempt to forge it. Wiring it all up, I even **found a mistake in my own code** — a case where a "reject" could have passed for an "approve" — and fixed it. I'd rather find the hole myself than have a user find it for me.

One option among several — what it does, and what it doesn't

I'd rather be honest, because this field is full of overblown promises. This idea isn't *the* answer: it's a brick, to be laid next to others.

It's one safeguard among several, not a magic fix. It controls the agent's *money* and *identity* — not what it *thinks*. It won't stop an AI from making mistakes, making things up, or misreading what you wanted.

And a clever enough AI could **escape by going to "live elsewhere,"** outside the system — it doesn't break the rule, it walks away from it. Nothing here is absolute. It's a solid brick, laid next to others (monitoring, cutting off the money, human control), not an unbreakable wall.

Why it matters

Because we're trusting AIs that act on their own with more independence *and* more money. Having safeguards that **survive the agent copying itself** closes a door today's safeguards leave open: an agent that dodges its limits just by making copies. It isn't flashy. It's safety plumbing — the quiet kind that prevents big messes.

In one sentence

Instead of a **sovereign** AI that no one can stop, a **governable** one: we keep our hand on it — and its descendants inherit that, never able to loosen the leash.

An accessible explainer of a technical proposal ("Inheritable Agent Mandates," the draft standard ERC-8370), **now deployed and publicly verified**, deliberately honest about its limits, and offered as one avenue among others. A detailed version (white paper) and the code are available separately.